

Ejercicio 1 parcial 2

Carnet : 00018824; A: 12; B: 99; Julio Alejandro Flores Diaz

Red : 10.12.99.0/24

```

Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
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VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 10.12.99.2 255.255.255.192 10.12.99.1
Checking for duplicate address...
save
show ipPC1 : 10.12.99.2 255.255.255.192 gateway 10.12.99.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.12.99.2/26
GATEWAY    : 10.12.99.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10020
RHOST:PORT : 127.0.0.1:10029
MTU        : 1500

PC1> trace 10.12.99.114
Trace to 10.12.99.114, 8 hops max, press Ctrl+C to stop
 1  10.12.99.1  45.792 ms 13.168 ms 16.656 ms
 2  10.12.99.122 45.953 ms 47.620 ms 46.801 ms
 3  *10.12.99.126 93.667 ms 79.893 ms
 4  **10.12.99.114 76.806 ms (ICMP type:3, code:3, Destination port unreachable)

PC1>

```

```

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Executing the startup file

PC2> ip 10.12.99.66 255.255.255.224 10.12.99.65
Checking for duplicate address...
save
show ip
PC1 : 10.12.99.66 255.255.255.224 gateway 10.12.99.65

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 10.12.99.66/27
GATEWAY    : 10.12.99.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10030
RHOST:PORT : 127.0.0.1:10031
MTU        : 1500

PC2> ping 10.12.99.98
10.12.99.98 icmp_seq=1 timeout
84 bytes from 10.12.99.98 icmp_seq=2 ttl=63 time=30.725 ms
84 bytes from 10.12.99.98 icmp_seq=3 ttl=63 time=30.958 ms
84 bytes from 10.12.99.98 icmp_seq=4 ttl=63 time=32.018 ms
84 bytes from 10.12.99.98 icmp_seq=5 ttl=63 time=31.966 ms

PC2>

```

PC2 PC1 PC3 PC4 R1 R2 R3 FPS N/A GPU CPU 27% NAT

```
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Press '?' to get help.

Executing the startup file

PC3> ip 10.12.99.98 255.255.255.240 10.12.99.97
Checking for duplicate address...
save
show ip
PC1 : 10.12.99.98 255.255.255.240 gateway 10.12.99.97

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> trace 10.12.99.114
trace to 10.12.99.114, 8 hops max, press Ctrl+C to stop
 1 10.12.99.97 14.768 ms 15.120 ms 15.959 ms
 2 10.12.99.122 46.212 ms 47.205 ms 46.423 ms
 3 10.12.99.126 77.243 ms 75.820 ms 77.877 ms
 4 *10.12.99.114 88.511 ms (ICMP type:3, code:3, Destination port unreachable)

PC3> show ip
NAME      : PC3[1]
IP/MASK   : 10.12.99.98/28
GATEWAY   : 10.12.99.97
DNS       :
MAC       : 00:50:79:66:68:02
I:PORT    : 10032
RHOST:PORT : 127.0.0.1:10033
MTU       : 1500

PC3> █
```

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27°C Parc soleado Buscar 10:42 5/6/2026 ENG 80%

PC2 PC1 PC3 PC4 R1 R2 R3 FPS N/A GPU CPU 13% NAT

```
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Press '?' to get help.

Executing the startup file

PC4> ip 10.12.99.114 255.255.255.248 10.12.99.113
Checking for duplicate address...
save
sJPC1 : 10.12.99.114 255.255.255.248 gateway 10.12.99.113

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> show ip
NAME      : PC4[1]
IP/MASK   : 10.12.99.114/29
GATEWAY   : 10.12.99.113
DNS       :
MAC       : 00:50:79:66:68:03
I:PORT    : 10034
RHOST:PORT : 127.0.0.1:10035
MTU       : 1500

PC4> trace 10.12.99.98
trace to 10.12.99.98, 8 hops max, press Ctrl+C to stop
 1 10.12.99.113 15.477 ms 15.268 ms 15.698 ms
 2 10.12.99.125 46.595 ms 48.767 ms 46.389 ms
 3 10.12.99.121 77.814 ms 79.688 ms 76.512 ms
 4 *10.12.99.98 91.080 ms (ICMP type:3, code:3, Destination port unreachable)

PC4> █
```

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27°C Parc soleado Buscar 10:42 5/6/2026 ENG 80%

```
PC2 PC1 PC3 PC4 R1 R2 R3 FPS N/A GPU CPU 11% DAT N/A

R1#
*Mar 1 00:04:52.059: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to down
R1#
*Mar 1 00:05:52.063: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R1#configure term
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip rou
R1(config)#ip route 10.12.99.124 255.255.255.252 10.12.99.122
R1(config)#ip route 10.12.99.112 255.255.255.248 10.12.99.122
R1(config)#end
R1#
Building configuration...

*Mar 1 00:08:31.103: %SYS-5-CONFIG_I: Configured from console by console[OK]
R1#show ip rou
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
C    10.12.99.96/28 is directly connected, FastEthernet1/0
S    10.12.99.112/29 [1/0] via 10.12.99.122
S    10.12.99.124/30 [1/0] via 10.12.99.122
C    10.12.99.120/30 is directly connected, Serial2/0
C    10.12.99.64/27 is directly connected, FastEthernet0/1
C    10.12.99.0/26 is directly connected, FastEthernet0/0
R1#show ip inter
R1#show ip interface br
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.12.99.1 YES manual up up
FastEthernet0/1 10.12.99.65 YES manual up up
FastEthernet1/0 10.12.99.97 YES manual up up
Serial2/0 10.12.99.121 YES manual up up
Serial2/1 unassigned YES unset administratively down down
Serial2/2 unassigned YES unset administratively down down
Serial2/3 unassigned YES unset administratively down down
R1#
```

```
PC2 PC1 PC3 PC4 R1 R2 R3 FPS N/A GPU CPU 15% DAT N/A

Serial2/3 unassigned YES unset administratively down down
R2#conf
R2#configure ter
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip rou
R2(config)#ip route 10.12.99.0 255.255.255.192 10.12.99.121
R2(config)#ip route 10.12.99.64 255.255.255.224 10.12.99.121
R2(config)#ip route 10.12.99.96 255.255.255.240 10.12.99.121
R2(config)#ip route 10.12.99.112 255.255.255.248 10.12.99.126
R2(config)#end
R2#
Building configuration...

*Mar 1 00:09:50.635: %SYS-5-CONFIG_I: Configured from console by console[OK]
R2#show ip rou
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    10.12.99.96/28 [1/0] via 10.12.99.121
S    10.12.99.112/29 [1/0] via 10.12.99.126
C    10.12.99.124/30 is directly connected, FastEthernet0/1
C    10.12.99.120/30 is directly connected, Serial2/0
S    10.12.99.64/27 [1/0] via 10.12.99.121
S    10.12.99.0/26 [1/0] via 10.12.99.121
R2#show ip inte
R2#show ip interface br
R2#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 unassigned YES unset administratively down down
FastEthernet0/1 10.12.99.125 YES manual up up
FastEthernet1/0 unassigned YES unset administratively down down
Serial2/0 10.12.99.122 YES manual up up
Serial2/1 unassigned YES unset administratively down down
Serial2/2 unassigned YES unset administratively down down
Serial2/3 unassigned YES unset administratively down down
R2#
```

PC2PC1PC3PC4R1R2R3

```
R3(config)#ip rou
R3(config)#ip route 10.12.99.0 255.255.255.192 10.12.99.125
R3(config)#ip route 10.12.99.64 255.255.255.224 10.12.99.125
R3(config)#ip route 10.12.99.96 255.255.255.240 10.12.99.125
R3(config)#show ip rou
R3(config)#show ip route
R3(config)#show ip route
^
% Invalid input detected at '^' marker.

R3(config)#end
R3#wr
Building configuration...

*Mar 1 00:12:31.727: %SYS-5-CONFIG_I: Configured from console by console[OK]
R3#show ip rou
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - OOR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    10.12.99.96/28 [1/0] via 10.12.99.125
C    10.12.99.112/29 is directly connected, FastEthernet0/0
C    10.12.99.124/30 is directly connected, FastEthernet0/1
S    10.12.99.120/30 [1/0] via 10.12.99.125
S    10.12.99.64/27 [1/0] via 10.12.99.125
S    10.12.99.0/26 [1/0] via 10.12.99.125
R3#show ip inter
R3#show ip interface br
R3#show ip interface brief
Interface              IP-Address      OK? Method Status              Protocol
FastEthernet0/0        10.12.99.113    YES manual up                    up
FastEthernet0/1        10.12.99.126    YES manual up                    up
FastEthernet1/0        unassigned      YES unset  administratively down down
Serial12/0             unassigned      YES unset  administratively down down
Serial12/1             unassigned      YES unset  administratively down down
Serial12/2             unassigned      YES unset  administratively down down
Serial12/3             unassigned      YES unset  administratively down down
R3#
```

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27°C
Parc. soleado

Buscar

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ENG 80% 10:42
5/6/2026

Dispositivo	Interface	IP	Máscara	Gateway
R1	Fe 0/0-P1	10.12.99.1	126	NA
R1	Fe 0/1-P2	10.12.99.65	127	
R1	Fe 1/0-P3	10.12.99.97	128	
R1	Serial 20-R2	10.12.99.121	130	

R2	Serial 1-R1	10.12.99.122	130	NA
R2	Fe 0/1-R3	10.12.99.123	130	

R3	Fe 0/1-R2	10.12.99.126	130	
R3	Fe 0/0-P4	10.12.99.113	129	

Pc1	H0/0-R1	10.12.99.2	126	10.12.99.1
Pc2	H0/0-R1	10.12.99.66	127	10.12.99.65
Pc3	H0/0-R1	10.12.99.98	128	10.12.99.97
Pc4	H0/0-R3	10.12.99.114	129	10.12.99.113

126 → ... 192 129 → ... 248

127 → ... 224

130 → ... 252

128 → ... 240

R1 — R2

R2 — R3 — Pc4

Carnet: 001011 88 124

$$A = 01 + 1 = 12$$

$$B = 88 + 1 = 99$$

10.02.89.0/24

PC	Host	PC	2	64	PC	2	32
PC1	50	PC1	2	64	PC2	2	32
PC2	25	PC2	6	126	PC3	5	127
PC3	12	PC3	12	255.255.255.192	PC4	2	255.255.255.224
PC4	6	PC4	24	16	PC5	2	255.255.255.240
R1-R2	2	R1-R2	4	128	R2-R3	2	130
R2-R3	2	R2-R3	2	240	R3-R4	2	252

PC	Rango	broadcast	mascara
PC1	10.02.89.0	10.02.89.1-10.02.89.63	255...192
PC2	10.02.89.64	10.02.89.64-10.02.89.127	255...224
PC3	10.02.89.128	10.02.89.128-10.02.89.191	255...240
PC4	10.02.89.192	10.02.89.192-10.02.89.255	255...252

PC	Rango	broadcast	mascara
PC1	10.02.89.0	10.02.89.1-10.02.89.63	255...192
PC2	10.02.89.64	10.02.89.64-10.02.89.127	255...224
PC3	10.02.89.128	10.02.89.128-10.02.89.191	255...240
PC4	10.02.89.192	10.02.89.192-10.02.89.255	255...252

PC	Rango	broadcast	mascara
PC1	10.02.89.0	10.02.89.1-10.02.89.63	255...192
PC2	10.02.89.64	10.02.89.64-10.02.89.127	255...224
PC3	10.02.89.128	10.02.89.128-10.02.89.191	255...240
PC4	10.02.89.192	10.02.89.192-10.02.89.255	255...252

PC	Rango	broadcast	mascara
PC1	10.02.89.0	10.02.89.1-10.02.89.63	255...192
PC2	10.02.89.64	10.02.89.64-10.02.89.127	255...224
PC3	10.02.89.128	10.02.89.128-10.02.89.191	255...240
PC4	10.02.89.192	10.02.89.192-10.02.89.255	255...252

PC	Rango	broadcast	mascara
PC1	10.02.89.0	10.02.89.1-10.02.89.63	255...192
PC2	10.02.89.64	10.02.89.64-10.02.89.127	255...224
PC3	10.02.89.128	10.02.89.128-10.02.89.191	255...240
PC4	10.02.89.192	10.02.89.192-10.02.89.255	255...252

PC	Rango	broadcast	mascara
PC1	10.02.89.0	10.02.89.1-10.02.89.63	255...192
PC2	10.02.89.64	10.02.89.64-10.02.89.127	255...224
PC3	10.02.89.128	10.02.89.128-10.02.89.191	255...240
PC4	10.02.89.192	10.02.89.192-10.02.89.255	255...252

PC	Rango	broadcast	mascara
PC1	10.02.89.0	10.02.89.1-10.02.89.63	255...192
PC2	10.02.89.64	10.02.89.64-10.02.89.127	255...224
PC3	10.02.89.128	10.02.89.128-10.02.89.191	255...240
PC4	10.02.89.192	10.02.89.192-10.02.89.255	255...252

Interfaz / Ruta, trace, ip